Aegis-Split

Escrow marketplace: clients fund milestones via PayPal, freelancers deliver on GitHub, and an AI auditor verifies work before payout.

1. The problem

Freelance and contract work often fails on trust. Clients worry about paying upfront and getting ghosted or low-quality deliverables. Freelancers worry about finishing work and never being paid — chasing invoices, disputing chargebacks, or learning that the client disappeared after handoff.

Traditional invoicing does not tie payment to proof of completion. Escrow helps, but manual escrow is slow and opaque. Aegis-Split automates the loop: agreed milestones, locked funds, verifiable delivery on GitHub, machine-assisted review against acceptance criteria, and release only when checks pass.

Who it helps: Product teams hiring contractors, agencies with milestone billing, and individual freelancers who want payment tied to objective delivery signals rather than email promises.

2. How escrow works

1. Agree — Client and freelancer create a project with milestones — task, acceptance criteria, and amount.

2. Verify — Both parties complete identity verification and monitoring consent (required before funding or linking code).

3. Fund — Client pays via PayPal Orders API. Funds are captured into the vault; money is locked, not paid out yet.

4. Build — Freelancer links a GitHub repository and pull request to the milestone, then merges when ready.

5. Audit — On merge, Aegis-Split fetches the PR diff, runs the AI auditor against acceptance criteria, and records pass or fail.

6. Pay — If the audit passes, PayPal Payouts releases that milestone's amount to the freelancer; both sides get notifications.

The same cycle repeats for each milestone until the project is complete. Funds never leave the vault for a milestone until its audit passes (or you intervene manually in demo/admin flows).

Exit, deposits & cancellation

Projects can end early with explicit money rules. Defaults: **early-exit penalty 50%** (client one-sided cancel) and a **freelancer security deposit** (default 10% of milestone totals, funded via PayPal Orders). Three paths:

- **Mutual** — request + accept; remaining vault returns to client; deposit returned to freelancer.
- **Client one-sided** — penalty share of remaining vault to freelancer; rest refunded; deposit returned.
- **Freelancer walk-away** — vault refunded to client; deposit forfeited to client.

Cancelled statuses (`CANCELLED_MUTUAL`, `CANCELLED_BY_CLIENT`, `ABANDONED_BY_FREELANCER`) block new funding, GitHub links, and merges. Exits claim an `ExitEvent` in the database before PayPal transfers to prevent double payouts. Users must agree to **Consent Agreement v2.0.0** (identity, platform-only comms, exit rules, conduct, disputes) with name and date.

Per-project text chat

Client and freelancer can exchange plain-text messages on each project via `GET/POST /api/projects/:projectId/messages`. Messages are stored in PostgreSQL (max 2000 characters, non-empty body). Participants only; polling from the demo UI. Text only — no voice or video calls.

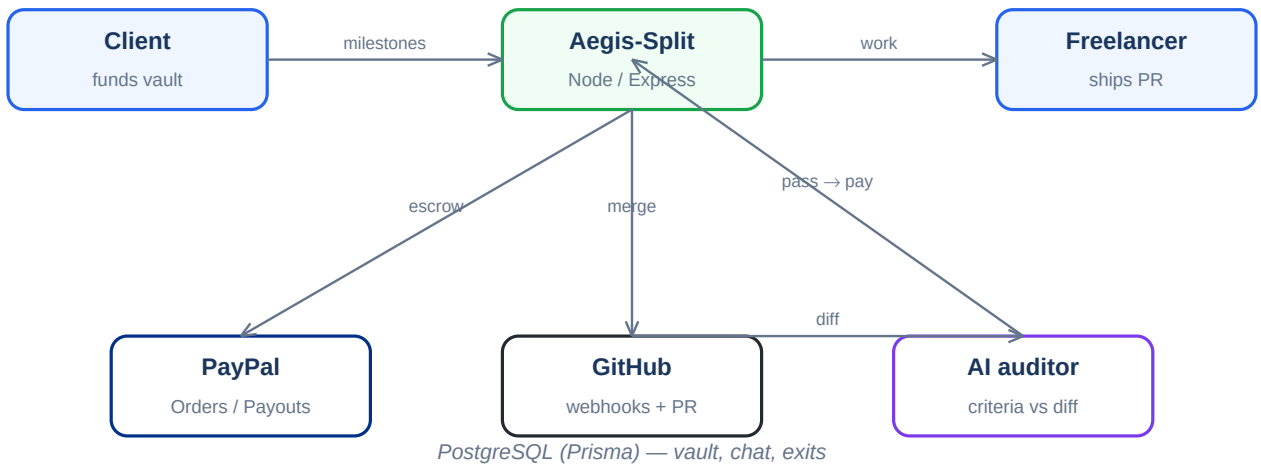
3. Tech stack

Runtime & API	Node.js 18+, Express — REST API and static demo UI on the same origin
Data	Prisma ORM + PostgreSQL (e.g. Neon on Render) — projects, vault, chat, exit events
Payments	PayPal REST — Orders API (fund / capture / deposit) and Payouts API (release)
Source control	GitHub — repo link per milestone, HMAC-verified pull_request webhooks
Verification	AI auditor — LLM when API keys are set; deterministic heuristics otherwise
Identity	DigiLocker / Stripe Identity scaffolds; consent logging with KYC gates
Safety	express-rate-limit on auth, vault, payouts, webhooks, admin demo; PayPal webhook signature verify
Hosting	Docker image; Render Blueprint (preferred), Railway, or Fly.io — live at aegis-split.onrender.com

4. Architecture

Browser clients talk to the Aegis-Split app. The app orchestrates PayPal escrow, listens for GitHub merge events, invokes the auditor, persists chat and exit state in PostgreSQL, and enforces gates before money moves.

System architecture (high level)



Key API areas: `/api/vault/*` (fund/capture), `/api/webhooks/github` + `/api/webhooks/paypal`, `/api/projects/*/milestones/*` (milestones, link, release), `/api/projects/*/messages` (chat), `/api/projects/*/exit/*` and cancel routes, `/api/notifications` (FUNDED, AUDIT_*, PAID, exit/deposit).

Trust, identity & safety hardening

Before creating projects, funding a vault, or linking production merges, each user must complete government-ID verification (DigiLocker for India, Stripe Identity scaffold elsewhere) and grant monitoring consent with a timestamped audit log. Middleware on protected routes returns `403` until both flags are set.

- **Rate limits** (PR #13) — per-IP windows on auth, vault fund/capture, milestone payout, GitHub/PayPal webhooks, and demo admin routes.
- **PayPal webhooks** — `POST /api/webhooks/paypal` verifies signatures via PayPal's REST verify API when `PAYPAL_WEBHOOK_ID` is set; optional backup capture path alongside browser return.
- **Write validation** — stricter validators on fund/capture, GitHub link, merge, consent, payouts, vault amounts (integer cents), and chat bodies.
- **Production errors** — generic 500 responses without leaking internal details.

Notifications: Idempotent in-app alerts fire for FUNDED, AUDIT_PASSED / AUDIT_FAILED, PAID, and exit/deposit events so both client and freelancer see the same timeline.

5. Live demo

Production sandbox deployment: **https://aegis-split.onrender.com**

The hosted app serves the demo UI at `/` and the Express API under `/api/*`. Use PayPal sandbox accounts and the demo seed flow to walk through fund → merge → audit → paid without installing locally.

6. How to test

Live (Render)

- Open https://aegis-split.onrender.com
- Click **Seed Sarah & Rohan** (uses demo admin key configured on the host)
- Switch user → complete gov ID verify + consent v2.0.0 (name + date)
- Create a project and milestone → **Fund vault** → approve in PayPal sandbox (or mock if enabled)
- Send a few **project chat** messages between client and freelancer
- Link GitHub repo → **Simulate merge** (or trigger a real webhook if the app URL is public)
- Confirm AI audit PASS, milestone PAID, and in-app notifications
- Optional: preview exit via `GET .../exit/preview?path=...` or mutual cancel

Local

```
cp .env.example .env && npm install && npx prisma migrate deploy && npm start
```

Open `http://localhost:3000`. Auth is stubbed: the demo UI sets `X-User-Id` when you pick a user.

Environment variables (high level — see `.env.example`):

`DATABASE_URL`	PostgreSQL connection string
`PAYPAL_CLIENT_ID / PAYPAL_CLIENT_SECRET`	Sandbox credentials (or PAYPAL_MOCK=true)
`PAYPAL_MODE`	sandbox
`PAYPAL_WEBHOOK_ID`	Required for verified PayPal webhook endpoint
`GITHUB_WEBHOOK_SECRET / GITHUB_TOKEN`	For real webhooks and PR diffs; optional for simulate-merge
`DEMO_ADMIN_KEY`	Defaults to hackathon-demo-key for seed and simulate-merge
`OPENAI_API_KEY or ANTHROPIC_API_KEY`	Optional — enables LLM auditor; otherwise deterministic

Run `npm test` for KYC, PayPal, GitHub, auditor, notification, chat, and exit-math unit checks. For full payout testing without PayPal credentials, set `PAYPAL_MOCK=true`.

API walkthrough (local or curl)

After seeding users, pick a user in the UI (sets `X-User-Id`) or pass the header manually:

```
POST /api/vault/fund { projectId, milestoneId, amountCents }

GET /api/vault/paypal/return?token=ORDER_ID (after PayPal approve)

GET/POST /api/projects/:projectId/messages

POST /api/projects/:projectId/milestones/:id/github { githubRepoFullName,
githubPrNumber }

POST /api/admin/demo/milestones/:id/simulate-merge (+ X-Demo-Admin-Key)

POST /api/projects/:projectId/milestones/:id/release { recipientEmail }
```

Real GitHub webhook: Expose your app URL (Render fulfills this), set `GITHUB_WEBHOOK_SECRET` on the repo webhook, and merge a linked PR. The handler verifies HMAC, pulls the diff with `GITHUB_TOKEN`, runs the auditor, and on PASS calls the same release path as simulate-merge.

PayPal sandbox notes: Default flow creates an order and returns an `approvalUrl` for the sandbox buyer. Capture runs on `/api/vault/paypal/return` when the order is APPROVED. Set the freelancer user's `paypal_email` to a sandbox personal account for payout testing.